

DecisionIQ Master Handbook

CHAPTER 4: The TechNova Global Case Study: Anatomy of a Supply Chain Breakdown

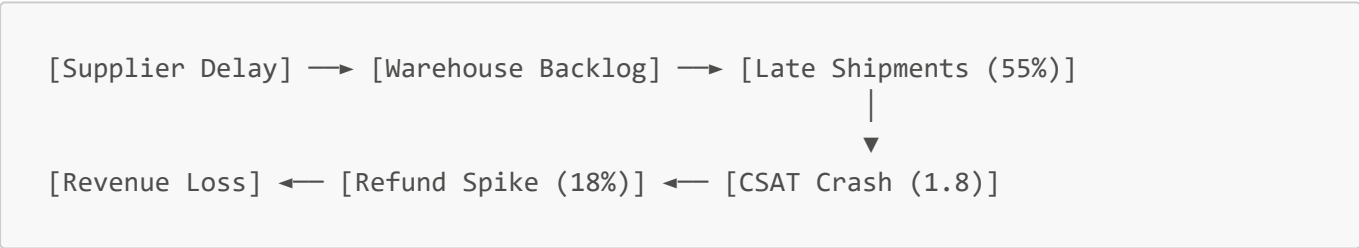
4.1 TechNova Global Profile

TechNova Global supplies cloud infrastructure and server hardware. It has multiple operational departments:

- **Sales:** Closes infrastructure and hardware orders.
- **Logistics & Warehouse:** Stores parts, manages safety stock, and ships servers.
- **Customer Support:** Resolves billing, technical, and shipping complaints.
- **Finance:** Invoices clients and processes refunds.

4.2 The Q2 2025 Failure Chain

To train our models and test our diagnostic SQL queries, we generate data containing a simulated **Q2 2025 systemic operational failure**:



1. **Supplier Delay:** Supplier Apex Technology Corp reliability drops from 95% to 72% due to chip shortages, delaying parts by 12 days.
2. **Warehouse Bottleneck:** The West Coast Fulfillment Center (**WH002**) falls behind. Compounded by understaffing, orders sit for an extra 7 days.
3. **Fulfillment Failure:** Delivery status for West Coast shipments becomes "Late" for 55% of hardware orders.
4. **Ticket Storm:** Customers complain about delays. CSAT crashes to 1.8.
5. **Financial Leakage:** Frustrated customers cancel orders, leading to an 18% refund spike and \$245,000 lost in region **REG01**.

4.3 Why this matters

By modeling a realistic cross-department failure chain rather than generating random numbers, our machine learning models can find correlations and predict customer churn based on logistical delays.